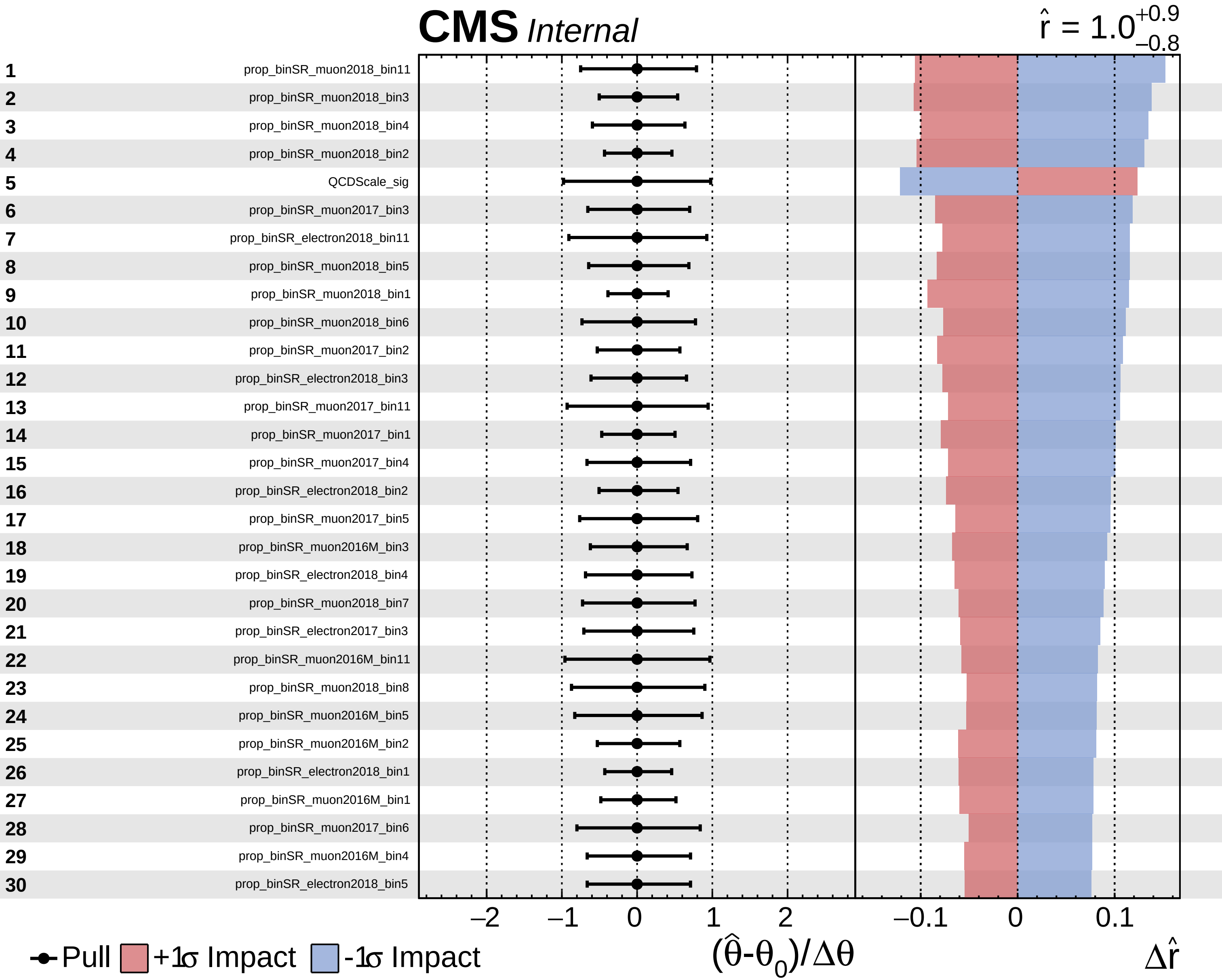
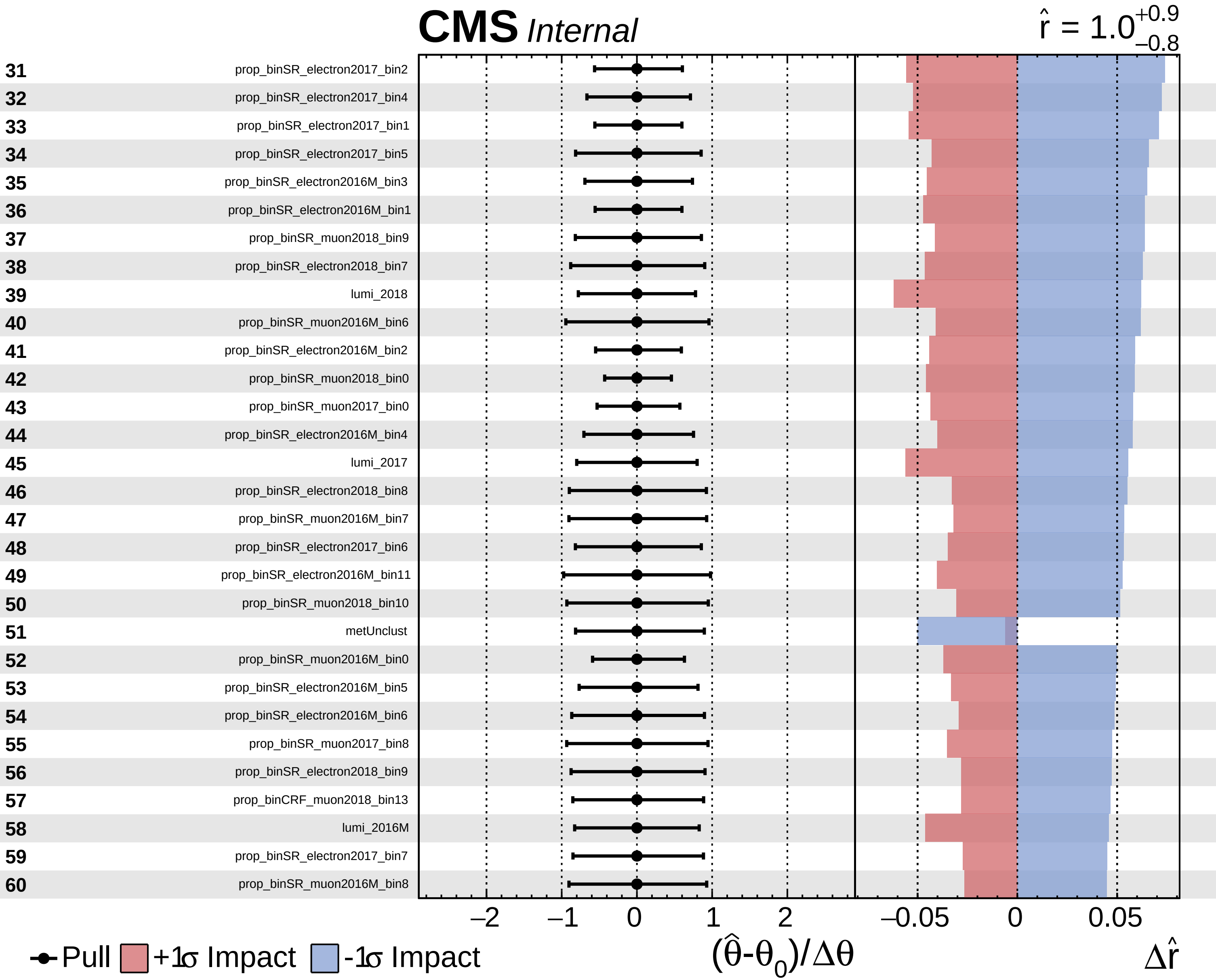
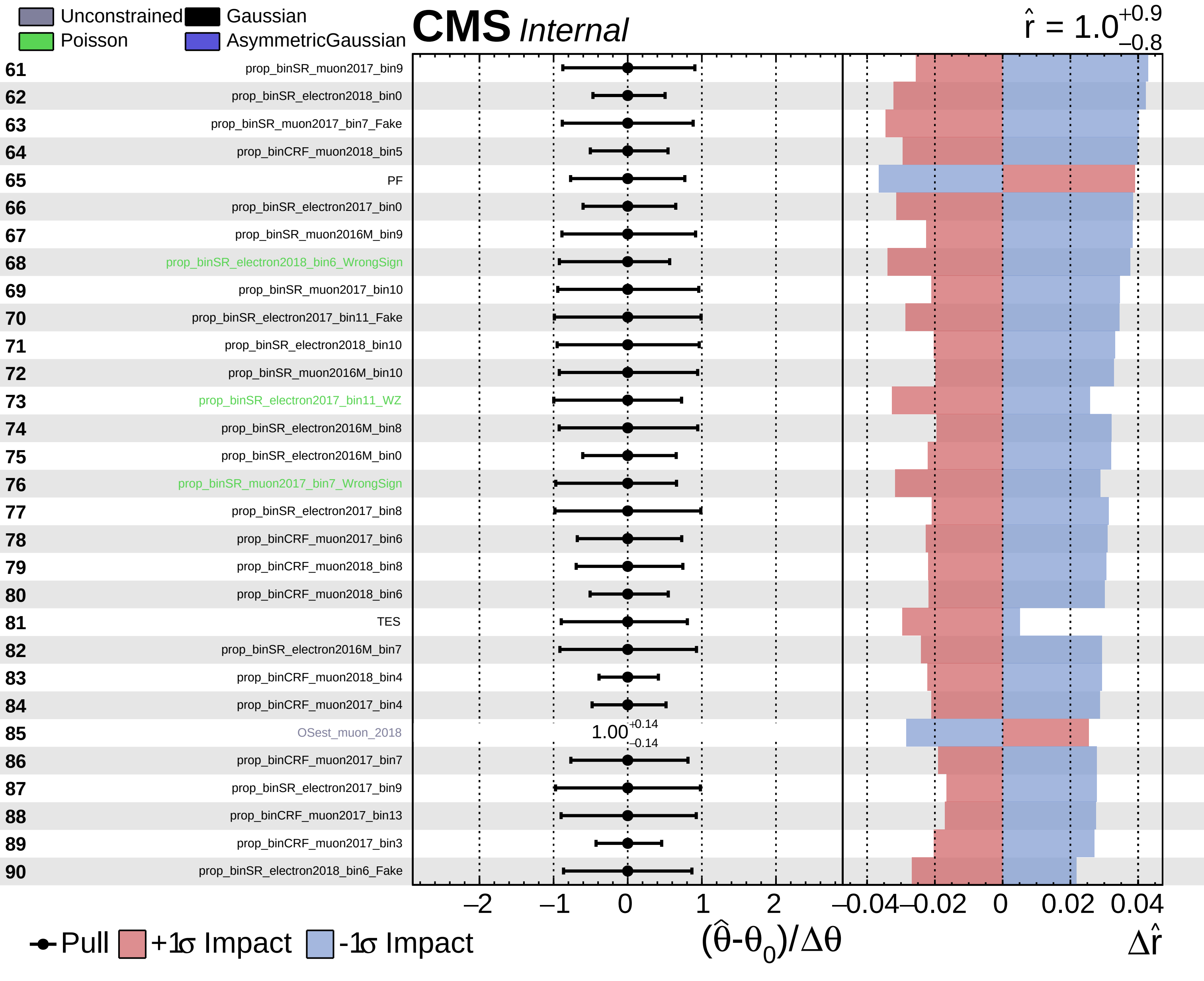
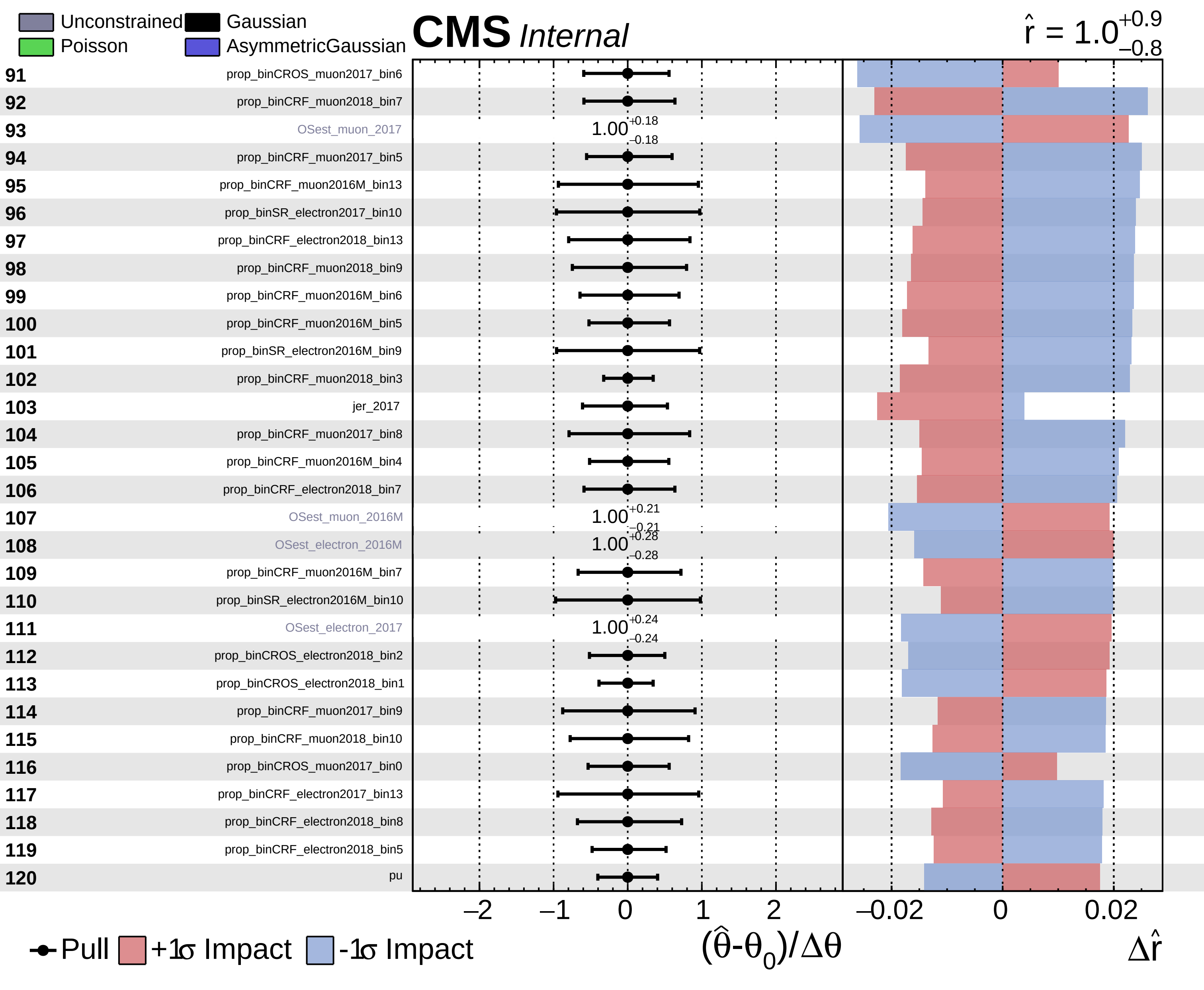

CMS Internal
r̂ = 1.0 +0.9 −0.8
1 prop_binSR_muon2018_bin11
2 prop_binSR_muon2018_bin3
3 prop_binSR_muon2018_bin4
4 prop_binSR_muon2018_bin2
5 QCDScale_sig
6 prop_binSR_muon2017_bin3
7 prop_binSR_electron2018_bin11
8 prop_binSR_muon2018_bin5
9 prop_binSR_muon2018_bin1
10 prop_binSR_muon2018_bin6
11 prop_binSR_muon2017_bin2
12 prop_binSR_electron2018_bin3
13 prop_binSR_muon2017_bin11
14 prop_binSR_muon2017_bin1
15 prop_binSR_muon2017_bin4
16 prop_binSR_electron2018_bin2
17 prop_binSR_muon2017_bin5
18 prop_binSR_muon2016M_bin3
19 prop_binSR_electron2018_bin4
20 prop_binSR_muon2018_bin7
21 prop_binSR_electron2017_bin3
22 prop_binSR_muon2016M_bin11
23 prop_binSR_muon2018_bin8
24 prop_binSR_muon2016M_bin5
25 prop_binSR_muon2016M_bin2
26 prop_binSR_electron2018_bin1
27 prop_binSR_muon2016M_bin1
28 prop_binSR_muon2017_bin6
29 prop_binSR_muon2016M_bin4
30 prop_binSR_electron2018_bin5
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

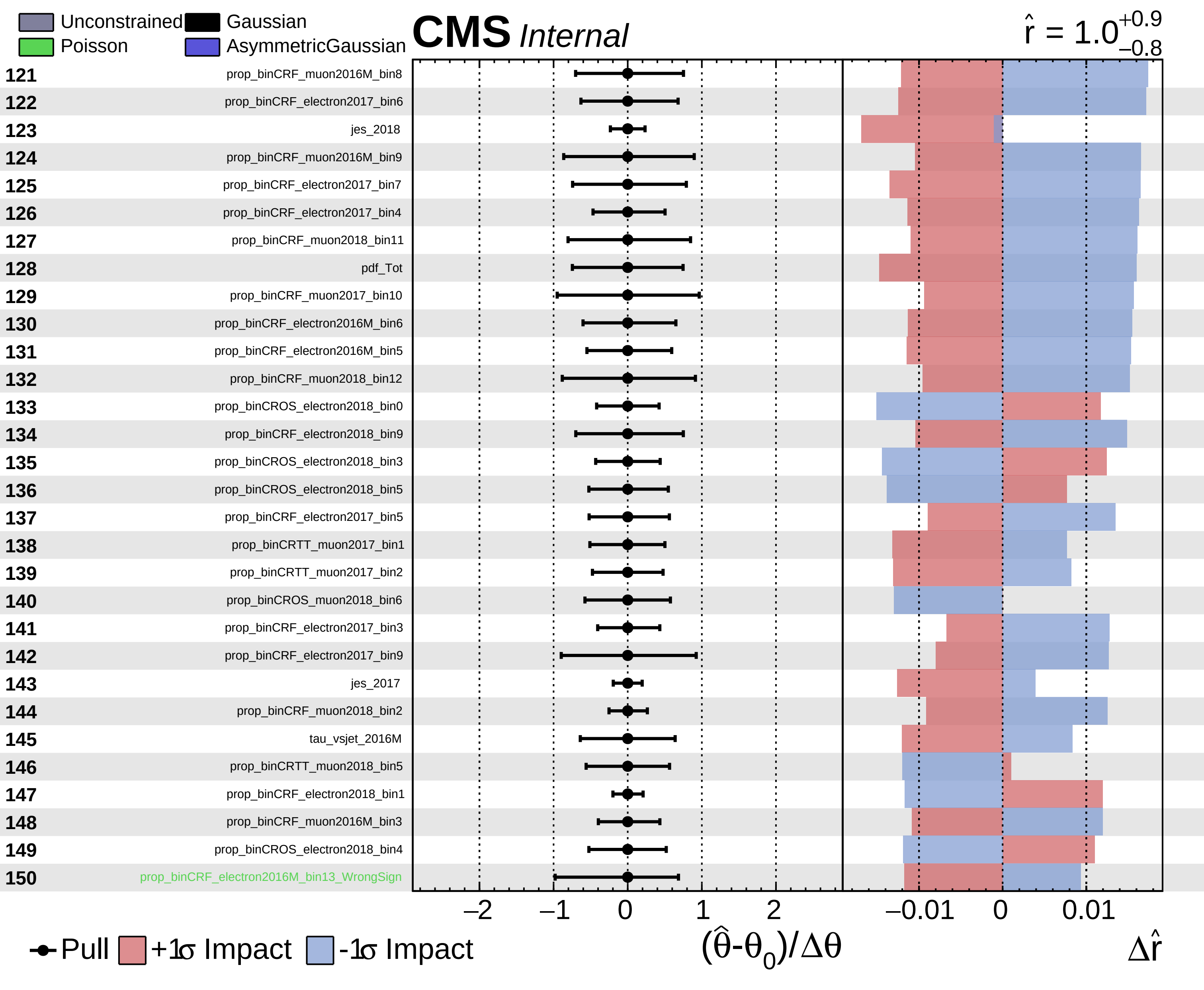


CMS Internal
r̂ = 1.0 +0.9 −0.8
31 prop_binSR_electron2017_bin2
32 prop_binSR_electron2017_bin4
33 prop_binSR_electron2017_bin1
34 prop_binSR_electron2017_bin5
35 prop_binSR_electron2016M_bin3
36 prop_binSR_electron2016M_bin1
37 prop_binSR_muon2018_bin9
38 prop_binSR_electron2018_bin7
39 lumi_2018
40 prop_binSR_muon2016M_bin6
41 prop_binSR_electron2016M_bin2
42 prop_binSR_muon2018_bin0
43 prop_binSR_muon2017_bin0
44 prop_binSR_electron2016M_bin4
45 lumi_2017
46 prop_binSR_electron2018_bin8
47 prop_binSR_muon2016M_bin7
48 prop_binSR_electron2017_bin6
49 prop_binSR_electron2016M_bin11
50 prop_binSR_muon2018_bin10
51 metUnclust
52 prop_binSR_muon2016M_bin0
53 prop_binSR_electron2016M_bin5
54 prop_binSR_electron2016M_bin6
55 prop_binSR_muon2017_bin8
56 prop_binSR_electron2018_bin9
57 prop_binCRF_muon2018_bin13
58 lumi_2016M
59 prop_binSR_electron2017_bin7
60 prop_binSR_muon2016M_bin8
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

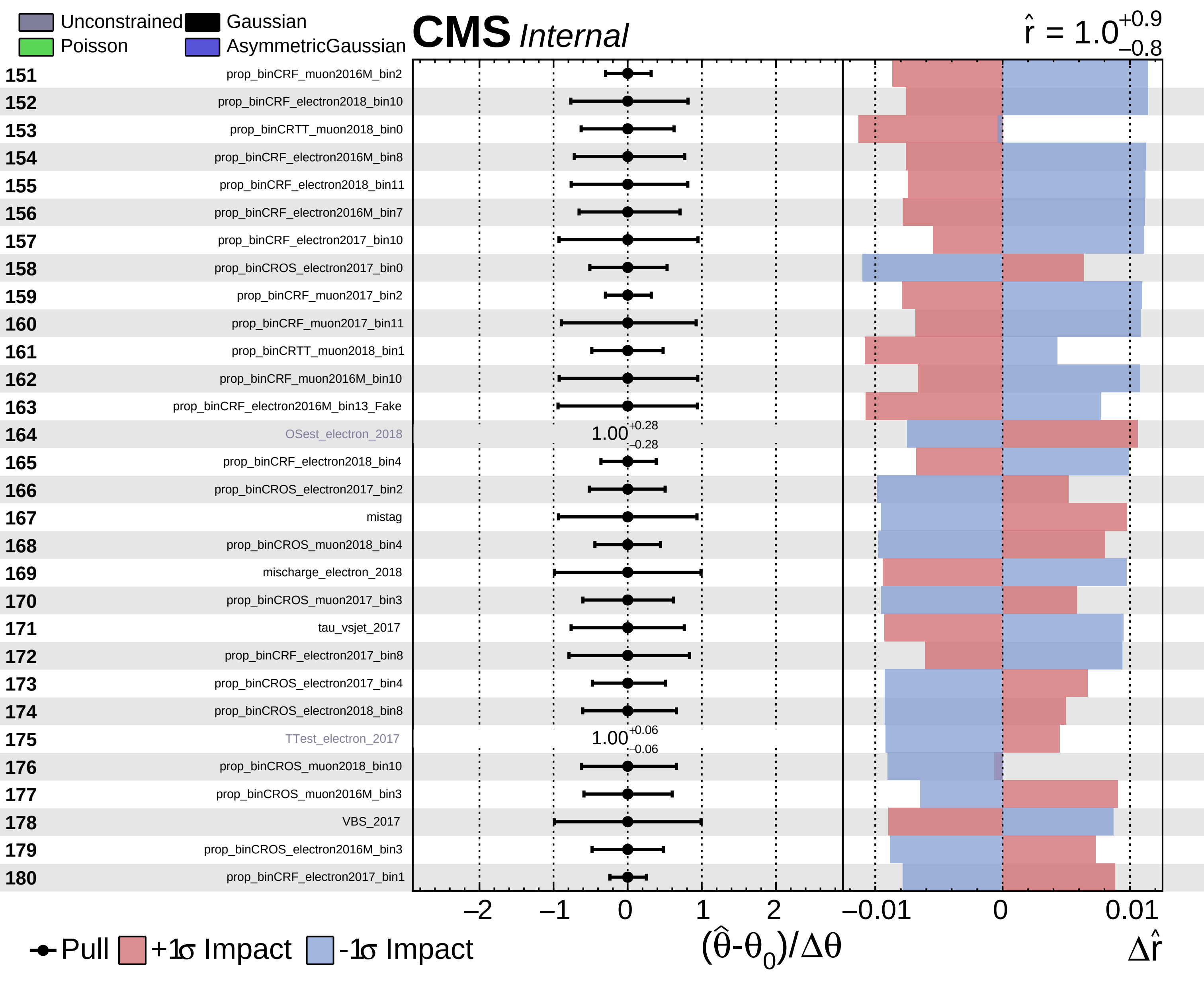

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
61 prop_binSR_muon2017_bin9
62 prop_binSR_electron2018_bin0
63 prop_binSR_muon2017_bin7_Fake
64 prop_binCRF_muon2018_bin5
65 PF
66 prop_binSR_electron2017_bin0
67 prop_binSR_muon2016M_bin9
68 prop_binSR_electron2018_bin6_WrongSign
69 prop_binSR_muon2017_bin10
70 prop_binSR_electron2017_bin11_Fake
71 prop_binSR_electron2018_bin10
72 prop_binSR_muon2016M_bin10
73 prop_binSR_electron2017_bin11_WZ
74 prop_binSR_electron2016M_bin8
75 prop_binSR_electron2016M_bin0
76 prop_binSR_muon2017_bin7_WrongSign
77 prop_binSR_electron2017_bin8
78 prop_binCRF_muon2017_bin6
79 prop_binCRF_muon2018_bin8
80 prop_binCRF_muon2018_bin6
81 TES
82 prop_binSR_electron2016M_bin7
83 prop_binCRF_muon2018_bin4
84 prop_binCRF_muon2017_bin4
85 OSest_muon_2018
$1.00^{+0.14}_{-0.14}$
86 prop_binCRF_muon2017_bin7
87 prop_binSR_electron2017_bin9
88 prop_binCRF_muon2017_bin13
89 prop_binCRF_muon2017_bin3
90 prop_binSR_electron2018_bin6_Fake
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

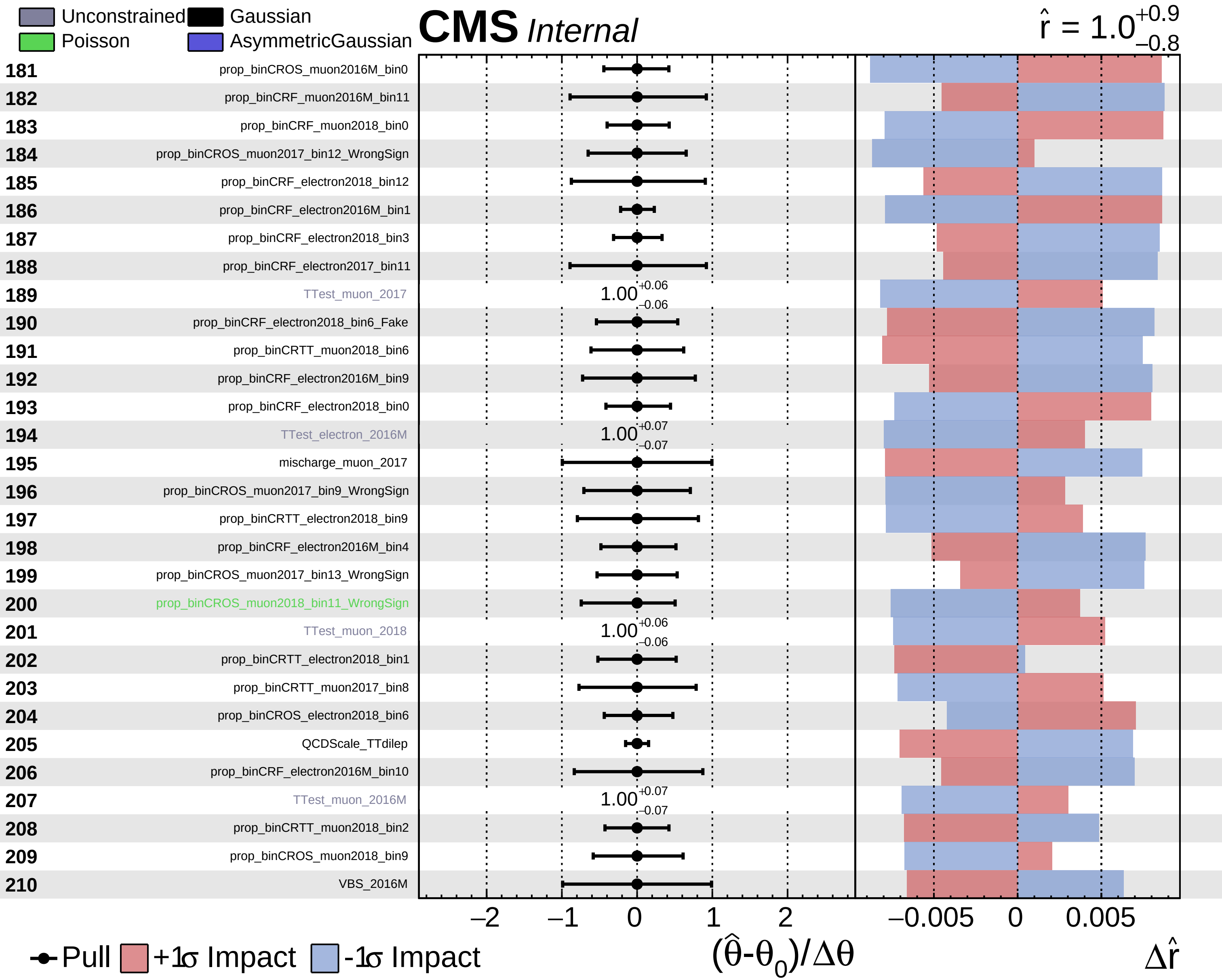


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
91 prop_binCROS_muon2017_bin6
92 prop_binCRF_muon2018_bin7
93 OSest_muon_2017 $1.00^{+0.18}_{-0.18}$
94 prop_binCRF_muon2017_bin5
95 prop_binCRF_muon2016M_bin13
96 prop_binSR_electron2017_bin10
97 prop_binCRF_electron2018_bin13
98 prop_binCRF_muon2018_bin9
99 prop_binCRF_muon2016M_bin6
100 prop_binCRF_muon2016M_bin5
101 prop_binSR_electron2016M_bin9
102 prop_binCRF_muon2018_bin3
103 jer_2017
104 prop_binCRF_muon2017_bin8
105 prop_binCRF_muon2016M_bin4
106 prop_binCRF_electron2018_bin7
107 OSest_muon_2016M $1.00^{+0.21}_{-0.21}$
108 OSest_electron_2016M $1.00^{+0.28}_{-0.28}$
109 prop_binCRF_muon2016M_bin7
110 prop_binSR_electron2016M_bin10
111 OSest_electron_2017 $1.00^{+0.24}_{-0.24}$
112 prop_binCROS_electron2018_bin2
113 prop_binCROS_electron2018_bin1
114 prop_binCRF_muon2017_bin9
115 prop_binCRF_muon2018_bin10
116 prop_binCROS_muon2017_bin0
117 prop_binCRF_electron2017_bin13
118 prop_binCRF_electron2018_bin8
119 prop_binCRF_electron2018_bin5
120 pu
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

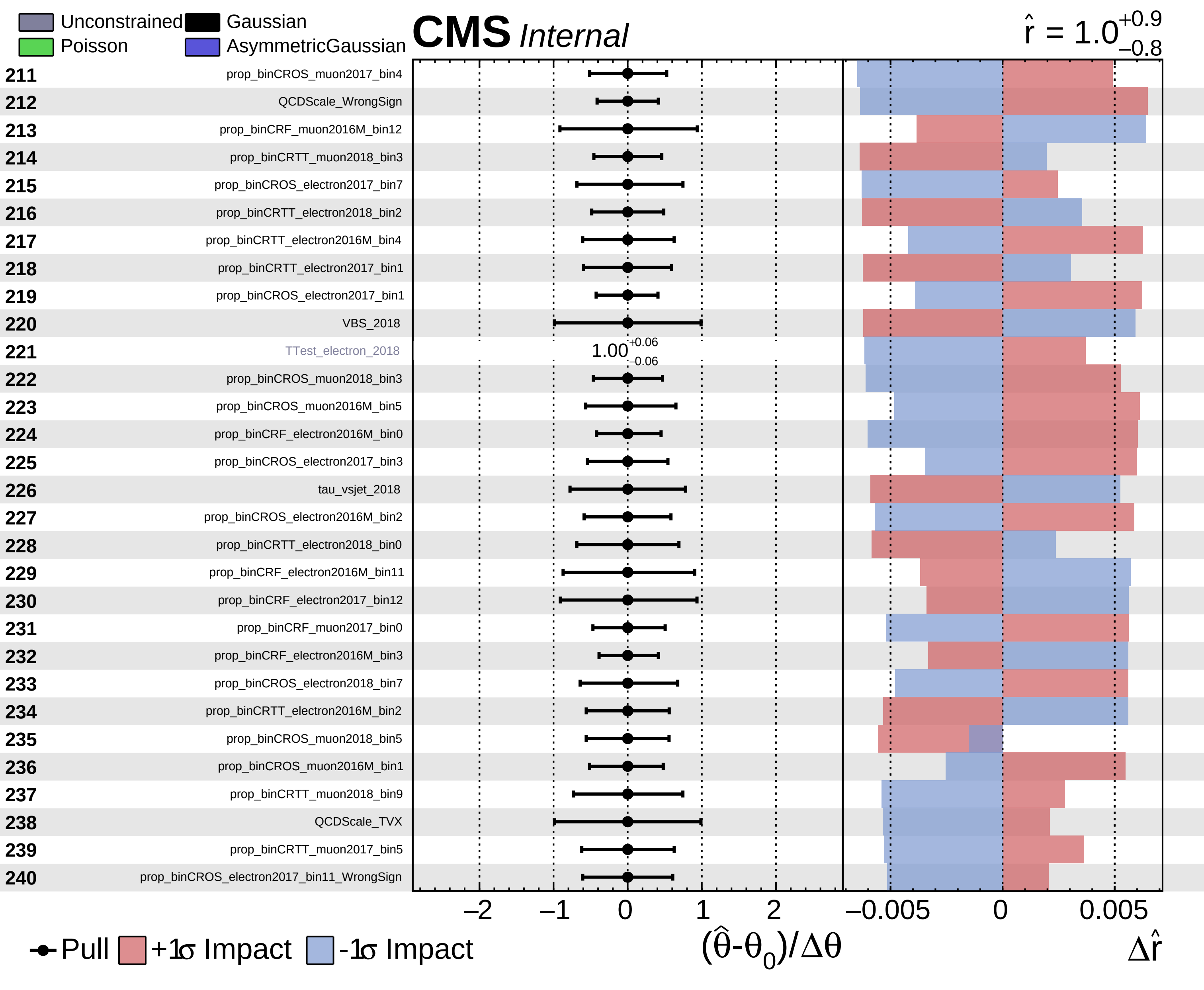


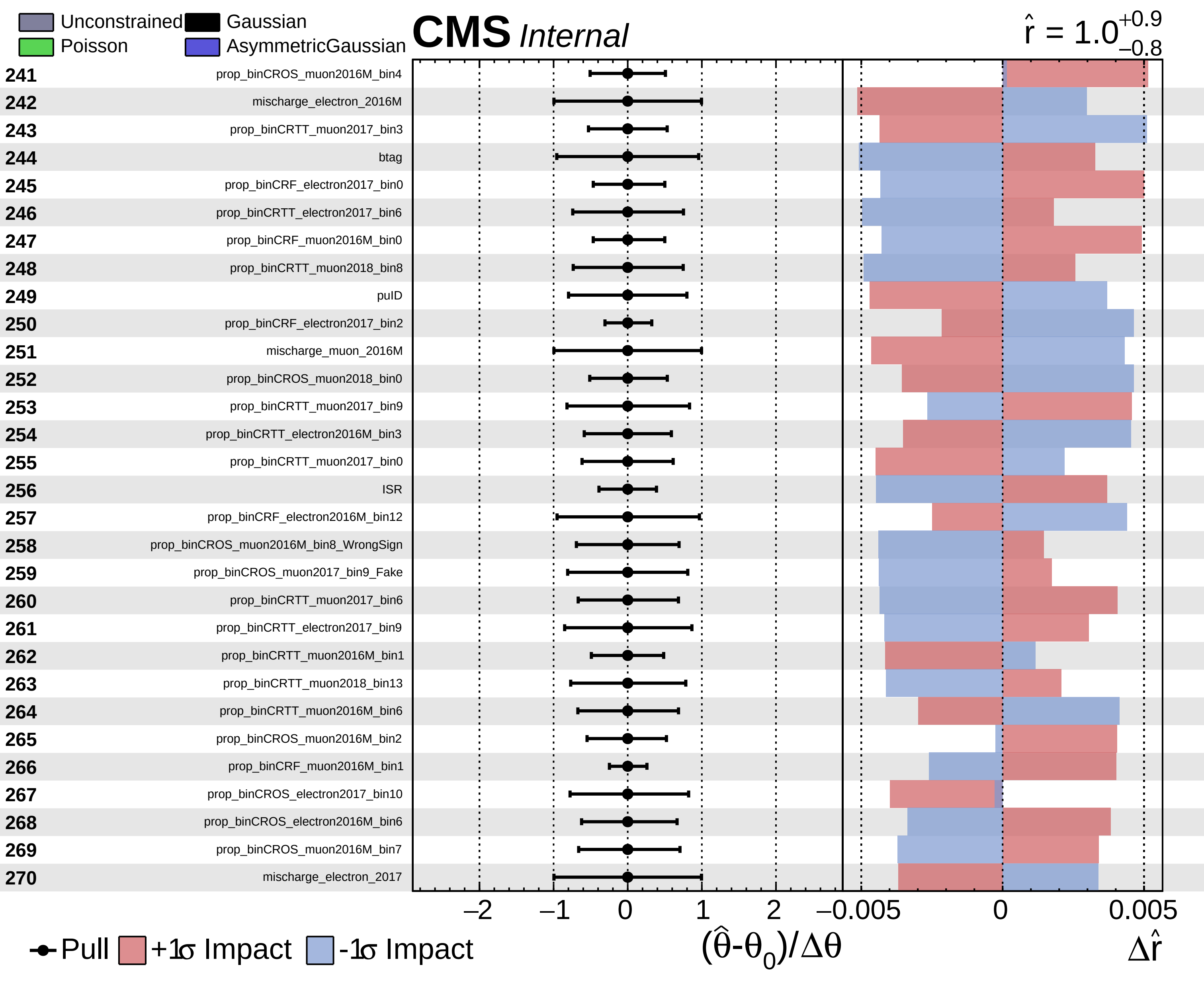
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
121 prop_binCRF_muon2016M_bin8
122 prop_binCRF_electron2017_bin6
123 jes_2018
124 prop_binCRF_muon2016M_bin9
125 prop_binCRF_electron2017_bin7
126 prop_binCRF_electron2017_bin4
127 prop_binCRF_muon2018_bin11
128 pdf_Tot
129 prop_binCRF_muon2017_bin10
130 prop_binCRF_electron2016M_bin6
131 prop_binCRF_electron2016M_bin5
132 prop_binCRF_muon2018_bin12
133 prop_binCROS_electron2018_bin0
134 prop_binCRF_electron2018_bin9
135 prop_binCROS_electron2018_bin3
136 prop_binCROS_electron2018_bin5
137 prop_binCRF_electron2017_bin5
138 prop_binCRTT_muon2017_bin1
139 prop_binCRTT_muon2017_bin2
140 prop_binCROS_muon2018_bin6
141 prop_binCRF_electron2017_bin3
142 prop_binCRF_electron2017_bin9
143 jes_2017
144 prop_binCRF_muon2018_bin2
145 tau_vsjet_2016M
146 prop_binCRTT_muon2018_bin5
147 prop_binCRF_electron2018_bin1
148 prop_binCRF_muon2016M_bin3
149 prop_binCROS_electron2018_bin4
150 prop_binCRF_electron2016M_bin13_WrongSign
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

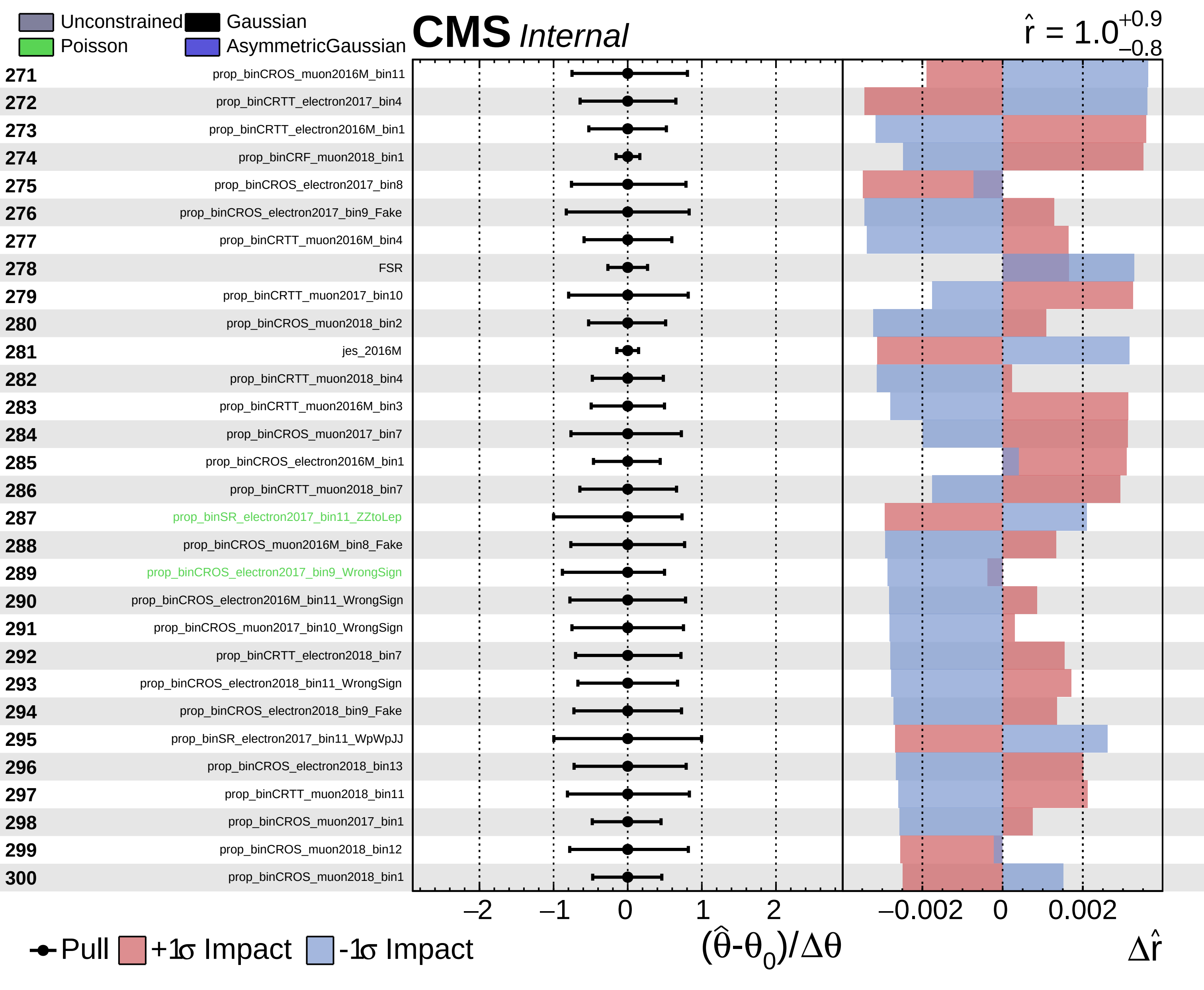

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
151 prop_binCRF_muon2016M_bin2
152 prop_binCRF_electron2018_bin10
153 prop_binCRTT_muon2018_bin0
154 prop_binCRF_electron2016M_bin8
155 prop_binCRF_electron2018_bin11
156 prop_binCRF_electron2016M_bin7
157 prop_binCRF_electron2017_bin10
158 prop_binCROS_electron2017_bin0
159 prop_binCRF_muon2017_bin2
160 prop_binCRF_muon2017_bin11
161 prop_binCRTT_muon2018_bin1
162 prop_binCRF_muon2016M_bin10
163 prop_binCRF_electron2016M_bin13_Fake
164 OSest_electron_2018 $1.00^{+0.28}_{-0.28}$
165 prop_binCRF_electron2018_bin4
166 prop_binCROS_electron2017_bin2
167 mistag
168 prop_binCROS_muon2018_bin4
169 mischarge_electron_2018
170 prop_binCROS_muon2017_bin3
171 tau_vsjet_2017
172 prop_binCRF_electron2017_bin8
173 prop_binCROS_electron2017_bin4
174 prop_binCROS_electron2018_bin8
175 TTest_electron_2017 $1.00^{+0.06}_{-0.06}$
176 prop_binCROS_muon2018_bin10
177 prop_binCROS_muon2016M_bin3
178 VBS_2017
179 prop_binCROS_electron2016M_bin3
180 prop_binCRF_electron2017_bin1
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

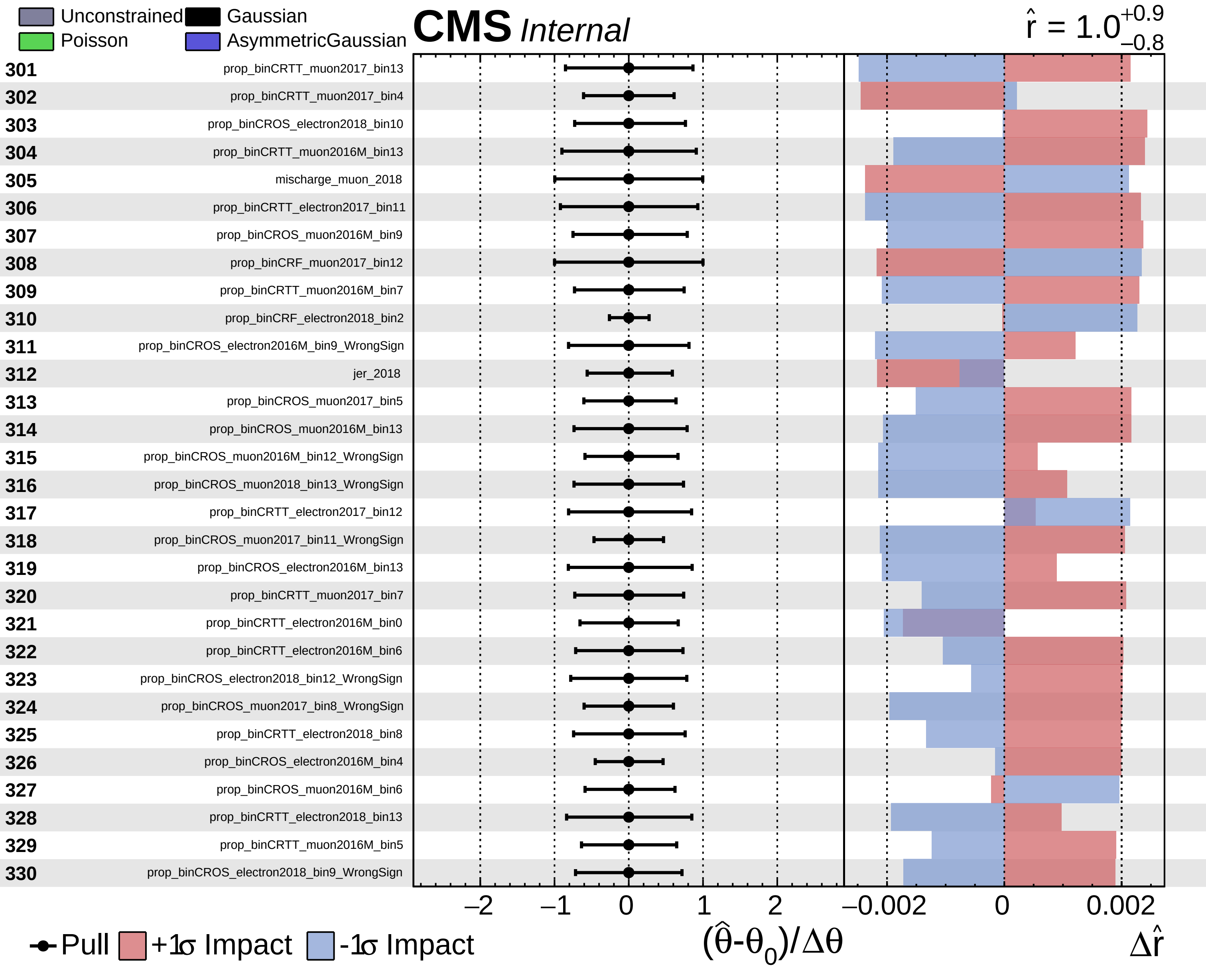


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
181 prop_binCROS_muon2016M_bin0
182 prop_binCRF_muon2016M_bin11
183 prop_binCRF_muon2018_bin0
184 prop_binCROS_muon2017_bin12_WrongSign
185 prop_binCRF_electron2018_bin12
186 prop_binCRF_electron2016M_bin1
187 prop_binCRF_electron2018_bin3
188 prop_binCRF_electron2017_bin11
189 TTest_muon_2017
$1.00^{+0.06}_{-0.06}$
190 prop_binCRF_electron2018_bin6_Fake
191 prop_binCRTT_muon2018_bin6
192 prop_binCRF_electron2016M_bin9
193 prop_binCRF_electron2018_bin0
194 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
195 mischarge_muon_2017
196 prop_binCROS_muon2017_bin9_WrongSign
197 prop_binCRTT_electron2018_bin9
198 prop_binCRF_electron2016M_bin4
199 prop_binCROS_muon2017_bin13_WrongSign
200 prop_binCROS_muon2018_bin11_WrongSign
201 TTest_muon_2018
$1.00^{+0.06}_{-0.06}$
202 prop_binCRTT_electron2018_bin1
203 prop_binCRTT_muon2017_bin8
204 prop_binCROS_electron2018_bin6
205 QCDScale_TTdilep
206 prop_binCRF_electron2016M_bin10
207 TTest_muon_2016M
$1.00^{+0.07}_{-0.07}$
208 prop_binCRTT_muon2018_bin2
209 prop_binCROS_muon2018_bin9
210 VBS_2016M
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

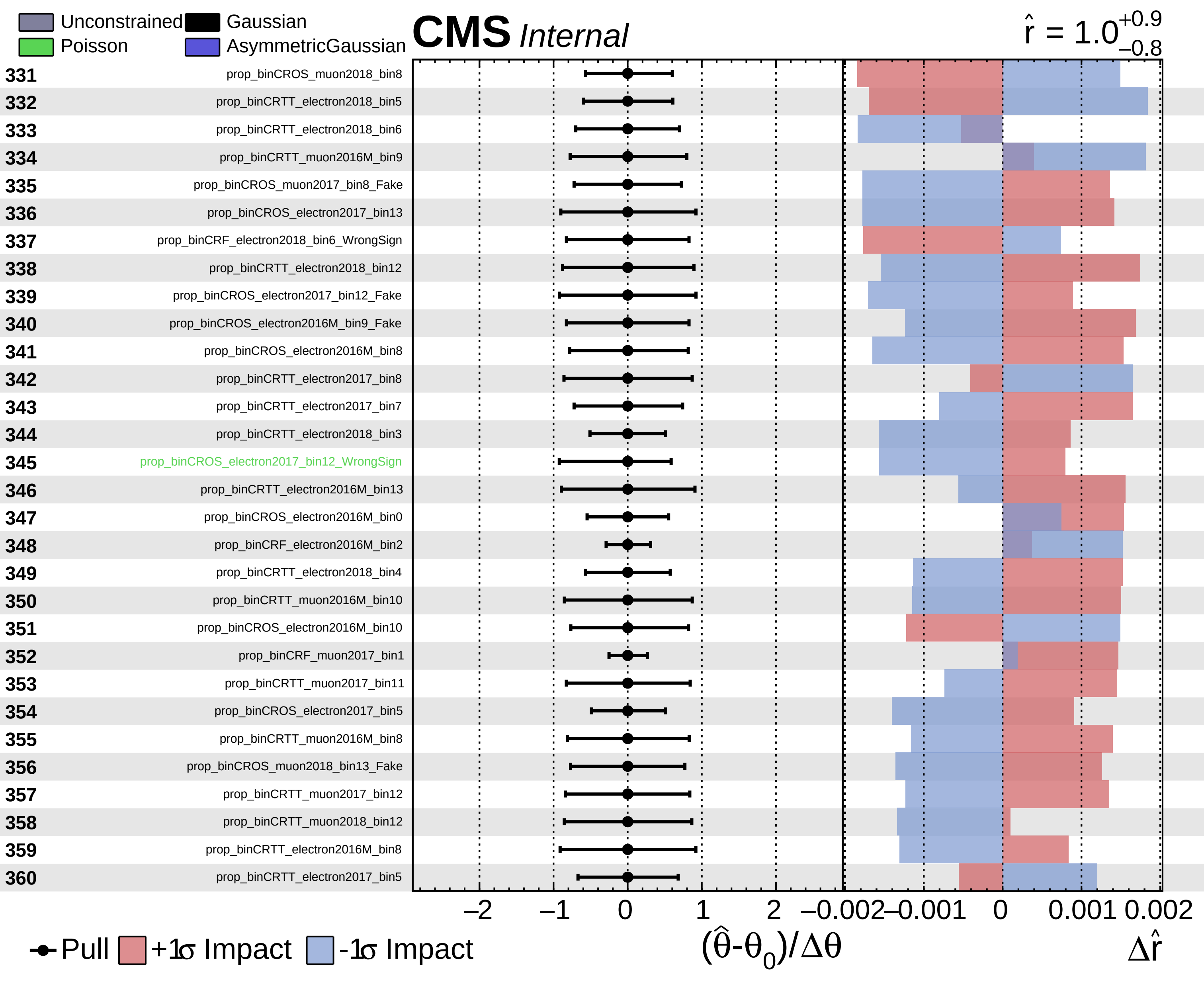

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.9 −0.8
211 prop_binCROS_muon2017_bin4
212 QCDScale_WrongSign
213 prop_binCRF_muon2016M_bin12
214 prop_binCRTT_muon2018_bin3
215 prop_binCROS_electron2017_bin7
216 prop_binCRTT_electron2018_bin2
217 prop_binCRTT_electron2016M_bin4
218 prop_binCRTT_electron2017_bin1
219 prop_binCROS_electron2017_bin1
220 VBS_2018
221 TTest_electron_2018
1.00 +0.06 −0.06
222 prop_binCROS_muon2018_bin3
223 prop_binCROS_muon2016M_bin5
224 prop_binCRF_electron2016M_bin0
225 prop_binCROS_electron2017_bin3
226 tau_vsjet_2018
227 prop_binCROS_electron2016M_bin2
228 prop_binCRTT_electron2018_bin0
229 prop_binCRF_electron2016M_bin11
230 prop_binCRF_electron2017_bin12
231 prop_binCRF_muon2017_bin0
232 prop_binCRF_electron2016M_bin3
233 prop_binCROS_electron2018_bin7
234 prop_binCRTT_electron2016M_bin2
235 prop_binCROS_muon2018_bin5
236 prop_binCROS_muon2016M_bin1
237 prop_binCRTT_muon2018_bin9
238 QCDScale_TVX
239 prop_binCRTT_muon2017_bin5
240 prop_binCROS_electron2017_bin11_WrongSign
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
241 prop_binCROS_muon2016M_bin4
242 mischarge_electron_2016M
243 prop_binCRTT_muon2017_bin3
244 btag
245 prop_binCRF_electron2017_bin0
246 prop_binCRTT_electron2017_bin6
247 prop_binCRF_muon2016M_bin0
248 prop_binCRTT_muon2018_bin8
249 puID
250 prop_binCRF_electron2017_bin2
251 mischarge_muon_2016M
252 prop_binCROS_muon2018_bin0
253 prop_binCRTT_muon2017_bin9
254 prop_binCRTT_electron2016M_bin3
255 prop_binCRTT_muon2017_bin0
256 ISR
257 prop_binCRF_electron2016M_bin12
258 prop_binCROS_muon2016M_bin8_WrongSign
259 prop_binCROS_muon2017_bin9_Fake
260 prop_binCRTT_muon2017_bin6
261 prop_binCRTT_electron2017_bin9
262 prop_binCRTT_muon2016M_bin1
263 prop_binCRTT_muon2018_bin13
264 prop_binCRTT_muon2016M_bin6
265 prop_binCROS_muon2016M_bin2
266 prop_binCRF_muon2016M_bin1
267 prop_binCROS_electron2017_bin10
268 prop_binCROS_electron2016M_bin6
269 prop_binCROS_muon2016M_bin7
270 mischarge_electron_2017
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

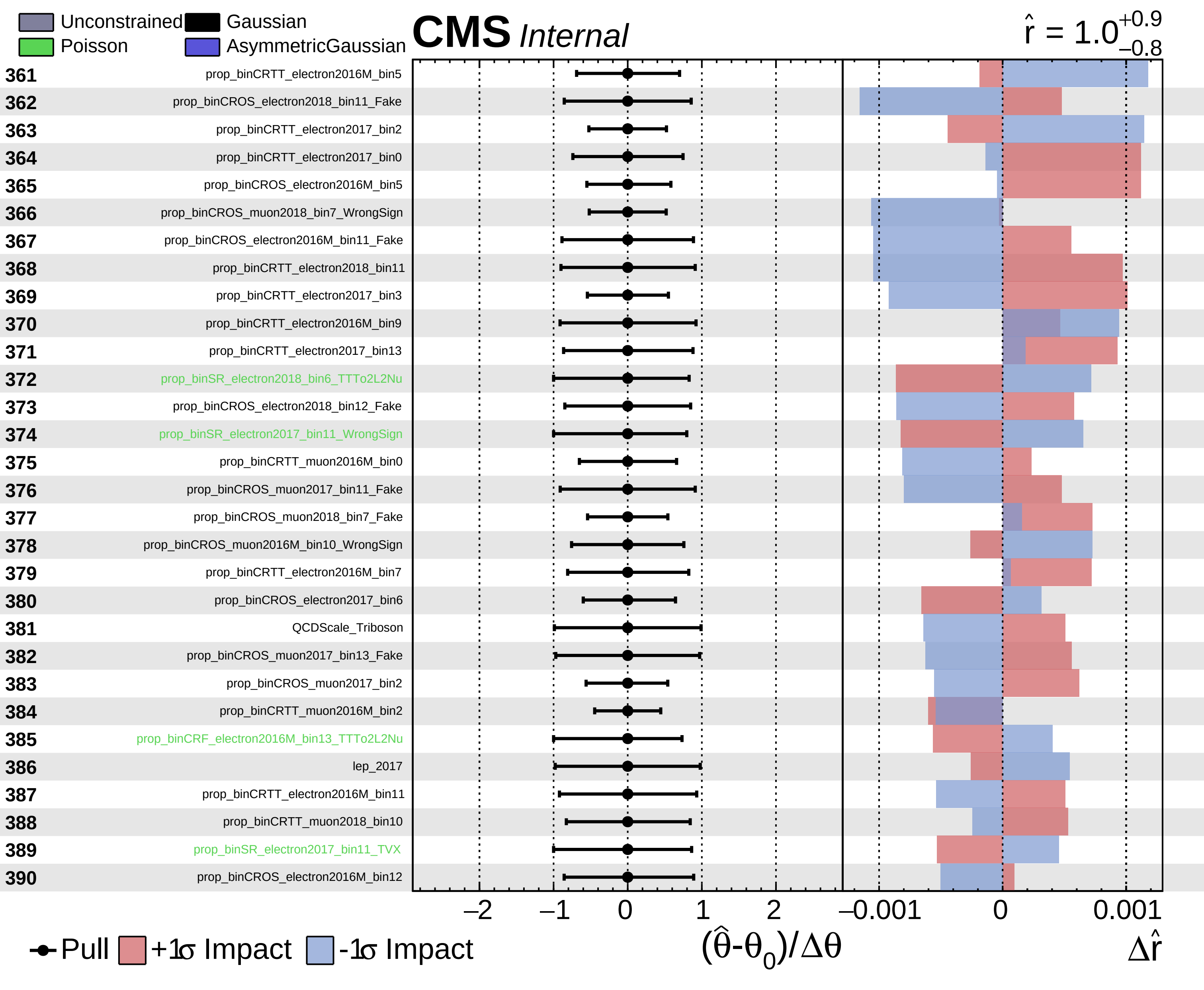

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
271 prop_binCROS_muon2016M_bin11
272 prop_binCRTT_electron2017_bin4
273 prop_binCRTT_electron2016M_bin1
274 prop_binCRF_muon2018_bin1
275 prop_binCROS_electron2017_bin8
276 prop_binCROS_electron2017_bin9_Fake
277 prop_binCRTT_muon2016M_bin4
278 FSR
279 prop_binCRTT_muon2017_bin10
280 prop_binCROS_muon2018_bin2
281 jes_2016M
282 prop_binCRTT_muon2018_bin4
283 prop_binCRTT_muon2016M_bin3
284 prop_binCROS_muon2017_bin7
285 prop_binCROS_electron2016M_bin1
286 prop_binCRTT_muon2018_bin7
287 prop_binSR_electron2017_bin11_ZZtoLep
288 prop_binCROS_muon2016M_bin8_Fake
289 prop_binCROS_electron2017_bin9_WrongSign
290 prop_binCROS_electron2016M_bin11_WrongSign
291 prop_binCROS_muon2017_bin10_WrongSign
292 prop_binCRTT_electron2018_bin7
293 prop_binCROS_electron2018_bin11_WrongSign
294 prop_binCROS_electron2018_bin9_Fake
295 prop_binSR_electron2017_bin11_WpWpJJ
296 prop_binCROS_electron2018_bin13
297 prop_binCRTT_muon2018_bin11
298 prop_binCROS_muon2017_bin1
299 prop_binCROS_muon2018_bin12
300 prop_binCROS_muon2018_bin1
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



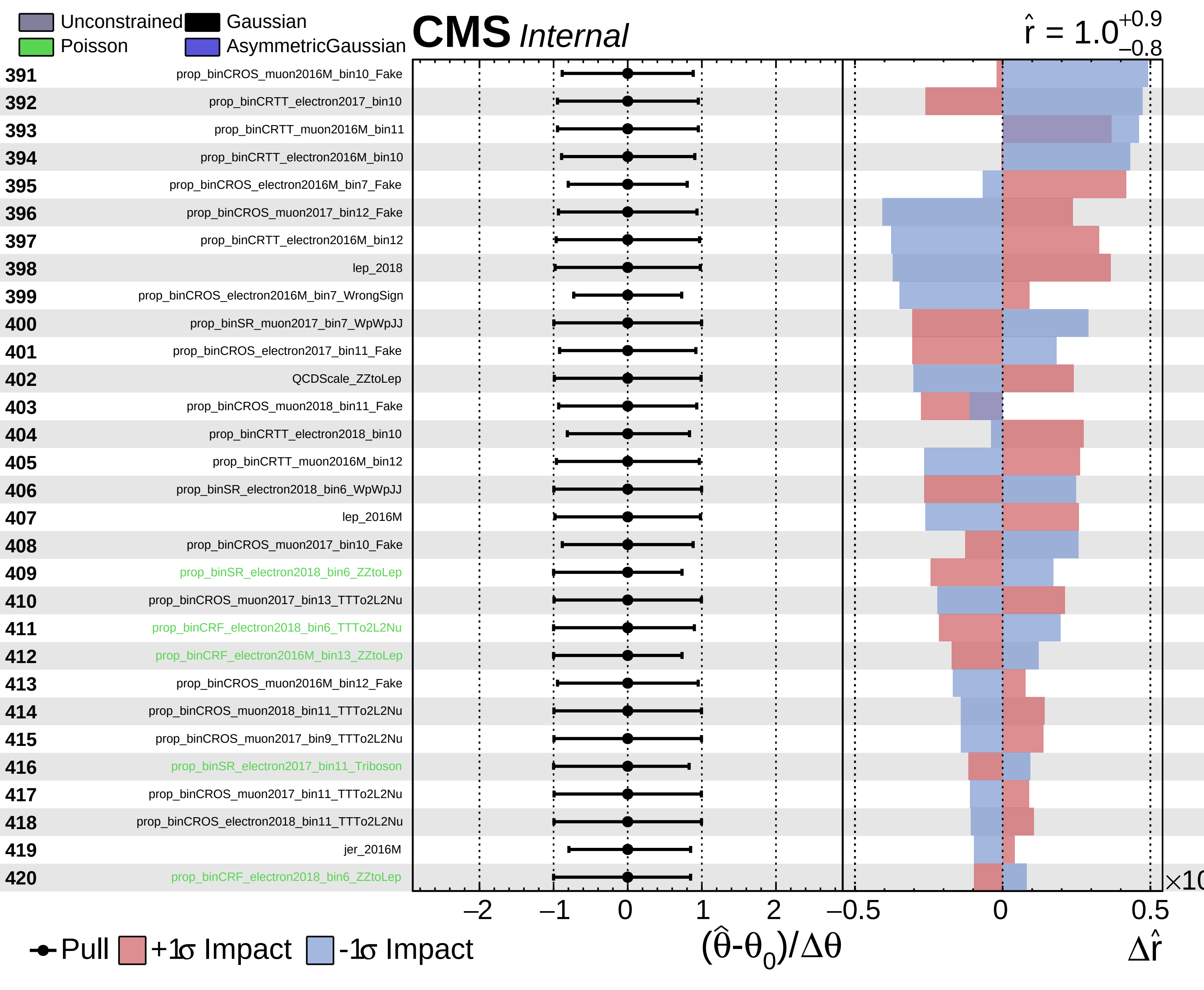
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
301 prop_binCRTT_muon2017_bin13
302 prop_binCRTT_muon2017_bin4
303 prop_binCROS_electron2018_bin10
304 prop_binCRTT_muon2016M_bin13
305 mischarge_muon_2018
306 prop_binCRTT_electron2017_bin11
307 prop_binCROS_muon2016M_bin9
308 prop_binCRF_muon2017_bin12
309 prop_binCRTT_muon2016M_bin7
310 prop_binCRF_electron2018_bin2
311 prop_binCROS_electron2016M_bin9_WrongSign
312 jer_2018
313 prop_binCROS_muon2017_bin5
314 prop_binCROS_muon2016M_bin13
315 prop_binCROS_muon2016M_bin12_WrongSign
316 prop_binCROS_muon2018_bin13_WrongSign
317 prop_binCRTT_electron2017_bin12
318 prop_binCROS_muon2017_bin11_WrongSign
319 prop_binCROS_electron2016M_bin13
320 prop_binCRTT_muon2017_bin7
321 prop_binCRTT_electron2016M_bin0
322 prop_binCRTT_electron2016M_bin6
323 prop_binCROS_electron2018_bin12_WrongSign
324 prop_binCROS_muon2017_bin8_WrongSign
325 prop_binCRTT_electron2018_bin8
326 prop_binCROS_electron2016M_bin4
327 prop_binCROS_muon2016M_bin6
328 prop_binCRTT_electron2018_bin13
329 prop_binCRTT_muon2016M_bin5
330 prop_binCROS_electron2018_bin9_WrongSign
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

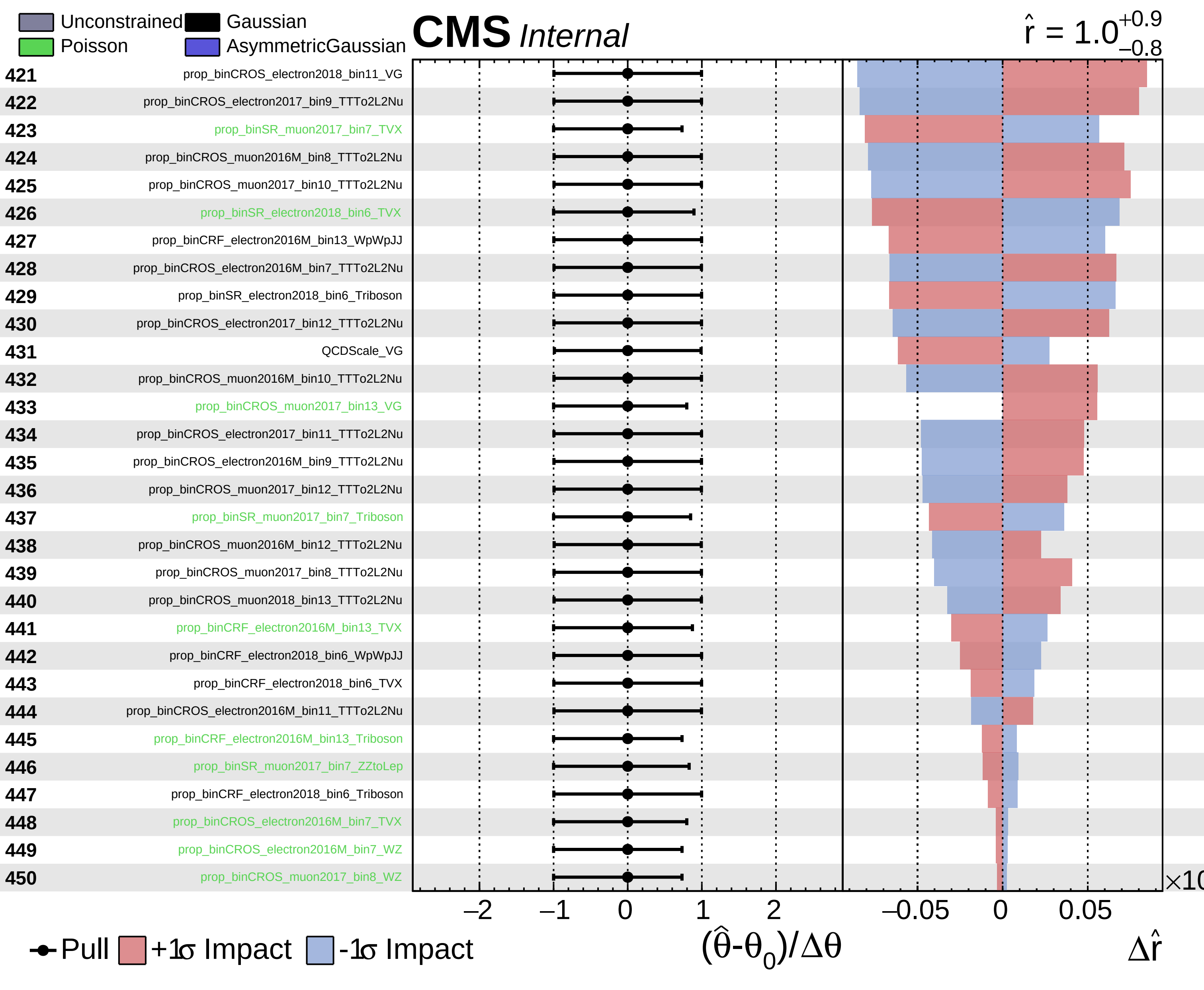


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
331 prop_binCROS_muon2018_bin8
332 prop_binCRTT_electron2018_bin5
333 prop_binCRTT_electron2018_bin6
334 prop_binCRTT_muon2016M_bin9
335 prop_binCROS_muon2017_bin8_Fake
336 prop_binCROS_electron2017_bin13
337 prop_binCRF_electron2018_bin6_WrongSign
338 prop_binCRTT_electron2018_bin12
339 prop_binCROS_electron2017_bin12_Fake
340 prop_binCROS_electron2016M_bin9_Fake
341 prop_binCROS_electron2016M_bin8
342 prop_binCRTT_electron2017_bin8
343 prop_binCRTT_electron2017_bin7
344 prop_binCRTT_electron2018_bin3
345 prop_binCROS_electron2017_bin12_WrongSign
346 prop_binCRTT_electron2016M_bin13
347 prop_binCROS_electron2016M_bin0
348 prop_binCRF_electron2016M_bin2
349 prop_binCRTT_electron2018_bin4
350 prop_binCRTT_muon2016M_bin10
351 prop_binCROS_electron2016M_bin10
352 prop_binCRF_muon2017_bin1
353 prop_binCRTT_muon2017_bin11
354 prop_binCROS_electron2017_bin5
355 prop_binCRTT_muon2016M_bin8
356 prop_binCROS_muon2018_bin13_Fake
357 prop_binCRTT_muon2017_bin12
358 prop_binCRTT_muon2018_bin12
359 prop_binCRTT_electron2016M_bin8
360 prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

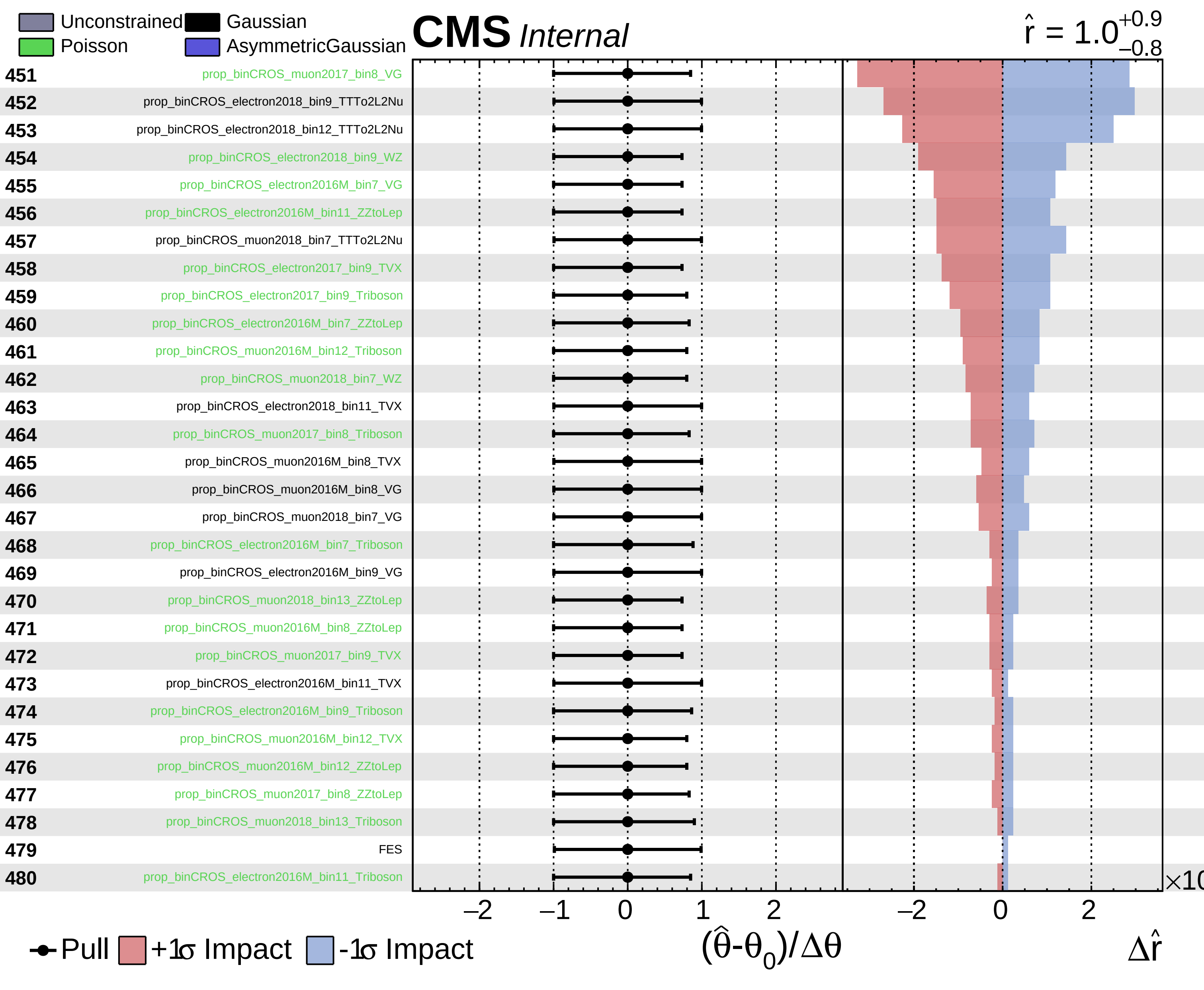


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
361 prop_binCRTT_electron2016M_bin5
362 prop_binCROS_electron2018_bin11_Fake
363 prop_binCRTT_electron2017_bin2
364 prop_binCRTT_electron2017_bin0
365 prop_binCROS_electron2016M_bin5
366 prop_binCROS_muon2018_bin7_WrongSign
367 prop_binCROS_electron2016M_bin11_Fake
368 prop_binCRTT_electron2018_bin11
369 prop_binCRTT_electron2017_bin3
370 prop_binCRTT_electron2016M_bin9
371 prop_binCRTT_electron2017_bin13
372 prop_binSR_electron2018_bin6_TTTo2L2Nu
373 prop_binCROS_electron2018_bin12_Fake
374 prop_binSR_electron2017_bin11_WrongSign
375 prop_binCRTT_muon2016M_bin0
376 prop_binCROS_muon2017_bin11_Fake
377 prop_binCROS_muon2018_bin7_Fake
378 prop_binCROS_muon2016M_bin10_WrongSign
379 prop_binCRTT_electron2016M_bin7
380 prop_binCROS_electron2017_bin6
381 QCDScale_Triboson
382 prop_binCROS_muon2017_bin13_Fake
383 prop_binCROS_muon2017_bin2
384 prop_binCRTT_muon2016M_bin2
385 prop_binCRF_electron2016M_bin13_TTTo2L2Nu
386 lep_2017
387 prop_binCRTT_electron2016M_bin11
388 prop_binCRTT_muon2018_bin10
389 prop_binSR_electron2017_bin11_TVX
390 prop_binCROS_electron2016M_bin12
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

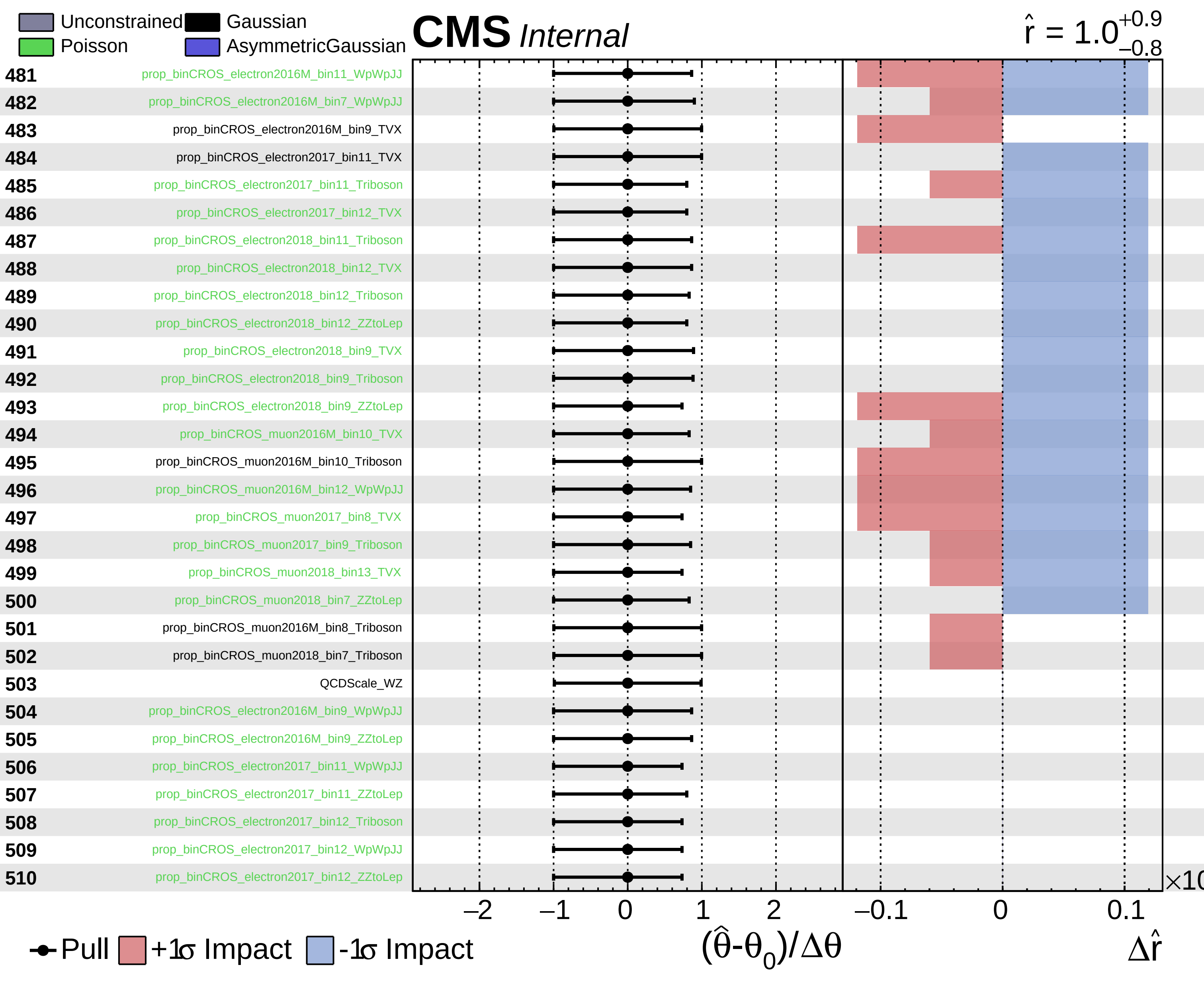

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
391 prop_binCROS_muon2016M_bin10_Fake
392 prop_binCRTT_electron2017_bin10
393 prop_binCRTT_muon2016M_bin11
394 prop_binCRTT_electron2016M_bin10
395 prop_binCROS_electron2016M_bin7_Fake
396 prop_binCROS_muon2017_bin12_Fake
397 prop_binCRTT_electron2016M_bin12
398 lep_2018
399 prop_binCROS_electron2016M_bin7_WrongSign
400 prop_binSR_muon2017_bin7_WpWpJJ
401 prop_binCROS_electron2017_bin11_Fake
402 QCDScale_ZZtoLep
403 prop_binCROS_muon2018_bin11_Fake
404 prop_binCRTT_electron2018_bin10
405 prop_binCRTT_muon2016M_bin12
406 prop_binSR_electron2018_bin6_WpWpJJ
407 lep_2016M
408 prop_binCROS_muon2017_bin10_Fake
409 prop_binSR_electron2018_bin6_ZZtoLep
410 prop_binCROS_muon2017_bin13_TTTo2L2Nu
411 prop_binCRF_electron2018_bin6_TTTo2L2Nu
412 prop_binCRF_electron2016M_bin13_ZZtoLep
413 prop_binCROS_muon2016M_bin12_Fake
414 prop_binCROS_muon2018_bin11_TTTo2L2Nu
415 prop_binCROS_muon2017_bin9_TTTo2L2Nu
416 prop_binSR_electron2017_bin11_Triboson
417 prop_binCROS_muon2017_bin11_TTTo2L2Nu
418 prop_binCROS_electron2018_bin11_TTTo2L2Nu
419 jer_2016M
420 prop_binCRF_electron2018_bin6_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



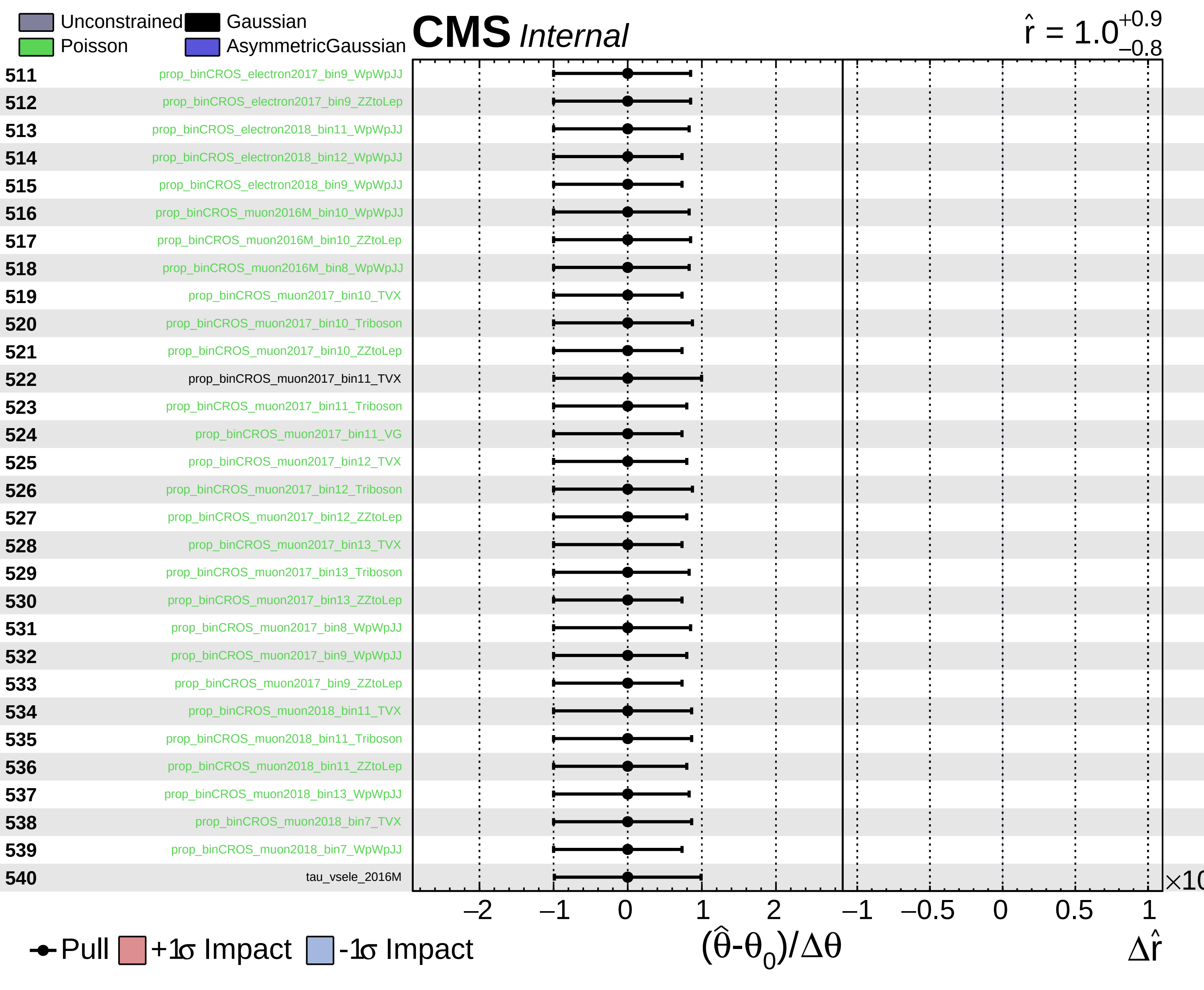
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
421 prop_binCROS_electron2018_bin11_VG
422 prop_binCROS_electron2017_bin9_TTTo2L2Nu
423 prop_binSR_muon2017_bin7_TVX
424 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
425 prop_binCROS_muon2017_bin10_TTTo2L2Nu
426 prop_binSR_electron2018_bin6_TVX
427 prop_binCRF_electron2016M_bin13_WpWpJJ
428 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
429 prop_binSR_electron2018_bin6_Triboson
430 prop_binCROS_electron2017_bin12_TTTo2L2Nu
431 QCDScale_VG
432 prop_binCROS_muon2016M_bin10_TTTo2L2Nu
433 prop_binCROS_muon2017_bin13_VG
434 prop_binCROS_electron2017_bin11_TTTo2L2Nu
435 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
436 prop_binCROS_muon2017_bin12_TTTo2L2Nu
437 prop_binSR_muon2017_bin7_Triboson
438 prop_binCROS_muon2016M_bin12_TTTo2L2Nu
439 prop_binCROS_muon2017_bin8_TTTo2L2Nu
440 prop_binCROS_muon2018_bin13_TTTo2L2Nu
441 prop_binCRF_electron2016M_bin13_TVX
442 prop_binCRF_electron2018_bin6_WpWpJJ
443 prop_binCRF_electron2018_bin6_TVX
444 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
445 prop_binCRF_electron2016M_bin13_Triboson
446 prop_binSR_muon2017_bin7_ZZtoLep
447 prop_binCRF_electron2018_bin6_Triboson
448 prop_binCROS_electron2016M_bin7_TVX
449 prop_binCROS_electron2016M_bin7_WZ
450 prop_binCROS_muon2017_bin8_WZ
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



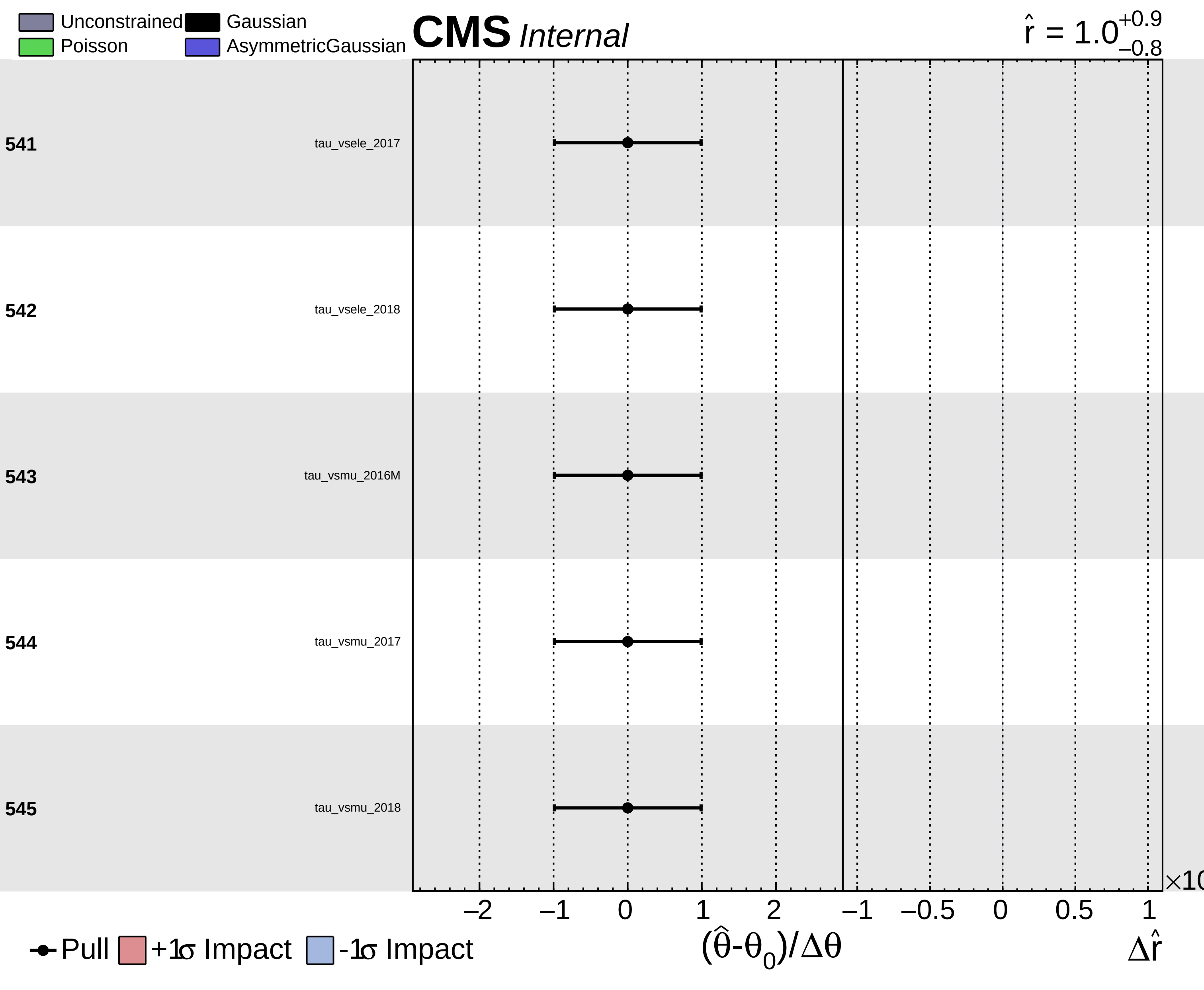
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.9 −0.8
451 prop_binCROS_muon2017_bin8_VG
452 prop_binCROS_electron2018_bin9_TTTo2L2Nu
453 prop_binCROS_electron2018_bin12_TTTo2L2Nu
454 prop_binCROS_electron2018_bin9_WZ
455 prop_binCROS_electron2016M_bin7_VG
456 prop_binCROS_electron2016M_bin11_ZZtoLep
457 prop_binCROS_muon2018_bin7_TTTo2L2Nu
458 prop_binCROS_electron2017_bin9_TVX
459 prop_binCROS_electron2017_bin9_Triboson
460 prop_binCROS_electron2016M_bin7_ZZtoLep
461 prop_binCROS_muon2016M_bin12_Triboson
462 prop_binCROS_muon2018_bin7_WZ
463 prop_binCROS_electron2018_bin11_TVX
464 prop_binCROS_muon2017_bin8_Triboson
465 prop_binCROS_muon2016M_bin8_TVX
466 prop_binCROS_muon2016M_bin8_VG
467 prop_binCROS_muon2018_bin7_VG
468 prop_binCROS_electron2016M_bin7_Triboson
469 prop_binCROS_electron2016M_bin9_VG
470 prop_binCROS_muon2018_bin13_ZZtoLep
471 prop_binCROS_muon2016M_bin8_ZZtoLep
472 prop_binCROS_muon2017_bin9_TVX
473 prop_binCROS_electron2016M_bin11_TVX
474 prop_binCROS_electron2016M_bin9_Triboson
475 prop_binCROS_muon2016M_bin12_TVX
476 prop_binCROS_muon2016M_bin12_ZZtoLep
477 prop_binCROS_muon2017_bin8_ZZtoLep
478 prop_binCROS_muon2018_bin13_Triboson
479 FES
480 prop_binCROS_electron2016M_bin11_Triboson
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
481 prop_binCROS_electron2016M_bin11_WpWpJJ
482 prop_binCROS_electron2016M_bin7_WpWpJJ
483 prop_binCROS_electron2016M_bin9_TVX
484 prop_binCROS_electron2017_bin11_TVX
485 prop_binCROS_electron2017_bin11_Triboson
486 prop_binCROS_electron2017_bin12_TVX
487 prop_binCROS_electron2018_bin11_Triboson
488 prop_binCROS_electron2018_bin12_TVX
489 prop_binCROS_electron2018_bin12_Triboson
490 prop_binCROS_electron2018_bin12_ZZtoLep
491 prop_binCROS_electron2018_bin9_TVX
492 prop_binCROS_electron2018_bin9_Triboson
493 prop_binCROS_electron2018_bin9_ZZtoLep
494 prop_binCROS_muon2016M_bin10_TVX
495 prop_binCROS_muon2016M_bin10_Triboson
496 prop_binCROS_muon2016M_bin12_WpWpJJ
497 prop_binCROS_muon2017_bin8_TVX
498 prop_binCROS_muon2017_bin9_Triboson
499 prop_binCROS_muon2018_bin13_TVX
500 prop_binCROS_muon2018_bin7_ZZtoLep
501 prop_binCROS_muon2016M_bin8_Triboson
502 prop_binCROS_muon2018_bin7_Triboson
503 QCDScale_WZ
504 prop_binCROS_electron2016M_bin9_WpWpJJ
505 prop_binCROS_electron2016M_bin9_ZZtoLep
506 prop_binCROS_electron2017_bin11_WpWpJJ
507 prop_binCROS_electron2017_bin11_ZZtoLep
508 prop_binCROS_electron2017_bin12_Triboson
509 prop_binCROS_electron2017_bin12_WpWpJJ
510 prop_binCROS_electron2017_bin12_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
511 prop_binCROS_electron2017_bin9_WpWpJJ
512 prop_binCROS_electron2017_bin9_ZZtoLep
513 prop_binCROS_electron2018_bin11_WpWpJJ
514 prop_binCROS_electron2018_bin12_WpWpJJ
515 prop_binCROS_electron2018_bin9_WpWpJJ
516 prop_binCROS_muon2016M_bin10_WpWpJJ
517 prop_binCROS_muon2016M_bin10_ZZtoLep
518 prop_binCROS_muon2016M_bin8_WpWpJJ
519 prop_binCROS_muon2017_bin10_TVX
520 prop_binCROS_muon2017_bin10_Triboson
521 prop_binCROS_muon2017_bin10_ZZtoLep
522 prop_binCROS_muon2017_bin11_TVX
523 prop_binCROS_muon2017_bin11_Triboson
524 prop_binCROS_muon2017_bin11_VG
525 prop_binCROS_muon2017_bin12_TVX
526 prop_binCROS_muon2017_bin12_Triboson
527 prop_binCROS_muon2017_bin12_ZZtoLep
528 prop_binCROS_muon2017_bin13_TVX
529 prop_binCROS_muon2017_bin13_Triboson
530 prop_binCROS_muon2017_bin13_ZZtoLep
531 prop_binCROS_muon2017_bin8_WpWpJJ
532 prop_binCROS_muon2017_bin9_WpWpJJ
533 prop_binCROS_muon2017_bin9_ZZtoLep
534 prop_binCROS_muon2018_bin11_TVX
535 prop_binCROS_muon2018_bin11_Triboson
536 prop_binCROS_muon2018_bin11_ZZtoLep
537 prop_binCROS_muon2018_bin13_WpWpJJ
538 prop_binCROS_muon2018_bin7_TVX
539 prop_binCROS_muon2018_bin7_WpWpJJ
540 tau_vsele_2016M
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
541
tau_vsele_2017
542
tau_vsele_2018
543
tau_vsmu_2016M
544
tau_vsmu_2017
545
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$